Vishal S

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Education

- **Sri Sivasubramaniya Nadar College of Engineering** — Chengalpattu, India
  B.E. in Computer Science and Engineering (CGPA 8.53) — *July 2023 – May 2027*
- **Indian Institute of Technology Madras** — Remote
  B.S. in Data Science and Applications (CGPA 8.25) — *July 2023 – May 2027*

Experience

- **Barclays Bank PLC** — Chennai, India
  BA4 Technology Summer Intern — *June – August 2026*
  - Designed and engineered a production automation tooling (Playwright, React, GitLab CI) from scratch to auto-generate branded client templates and created mobile builds (APKs and IPAs), achieving 88% fetch accuracy and 14-minute build times
- **Google Summer of Code (GSOC '26)** — Remote
  Contributor - German Center for Open Source AI — *June – September 2026*
  - Fixed anti-conservative p-values for smooth terms in pyGAM via Wood's (2013) Wald test (QR-based statistic, and a fractional chi square mixture) and validated to ∼1e-9 agreement with R's mgcv across 88 tests (PR #586)
- **Runverve - Startup** — Chennai, India
  Backend Developer Intern — *July – Sept, 2024*
  - Created APIs handling 1000+ daily requests, and designed efficient database schemas with Flask and Postgres

Research Publications

- **VQA in GI Images - MediaEval 2025**: Ranked 1st benchmarking PaliGemma, Florence-2, BLIP-2 on 159k QnA pairs with PEFT (LoRA), achieving BLEU 0.5, ROUGE 0.71
- **Multimodal Emotion Detection - SemEval 2025**: Ensemble of transformers with custom scoring (POS tagging, lexical rules, extremity scores), achieving 70% F1-score
- **Fake News Detection in Low Resource Languages - SPELLL 2024**: Classification system for Dravidian languages using different families of BERTs, achieving 96% avg. accuracy along with analysing impact of sentiment extremeness

Projects and Hackathons

- **IBM Z Datathon '25 - Global 2nd Place (out of 1384 teams)**: Multi-model time-series pipeline for demand forecasting in public kitchens, achieving a 90% score (1-MAPE) with $750 cash prize and L1CC credits
- **Intelligent Document Processing Pipeline**: Scalable platform processing multilingual handwritten documents with OCR, vector search, alerting, and RAG-powered summarization and QA
- **Business Analytics - Forte Furniture**: Analyzed 3L+ rows of inventory/sales data; ABC classification and S-ARIMA forecasting projected up to 10% reduction in holding costs
- **Efficient Graph Transformers**: Learned discrete token-selection to cut quadratic attention cost in Graph Transformers, with multiple ablations, benchmarked on mesh-based PDE surrogates
- **Additional**: SAP Hackfest National Qualifier (Ethical AI) ; Full-stack quiz app (Celery, Redis) ; Pac-Man with BFS multi-goal agents

Skills

- **AI & ML**: Deep Learning, NLP, Computer Vision, LLMs & RAG (LangChain, Hugging Face, FAISS)
- **Data Science**: Statistics, Probability, Time Series Analysis, Pandas, Matplotlib, Seaborn
- **Programming**: Python (NumPy, Scikit-learn, TensorFlow, PyTorch), Java, C, SQL, DSA, OOP
- **Tools & Web**: Docker, Git, GitLab CI, Spring Boot, Flask, Node, React, Vue, Redis, Celery

Extra Curricular Positions

- Department Association Event Organizer
- Coding Club Core Committee
- School Head Boy